Extreme Cold Weather Fuel Handler Handwear Survey

EXTREME COLD WEATHER (ECW) FUEL HANDLER (FH) 2-PIECE COVERALL AND HANDWEAR SURVEY

The purpose of this survey is to gather your opinions on the Extreme Cold Weather (ECW) Fuel Handler (FH) 2-Piece Coverall and Handwear that you have been evaluating. Project Officers at the U.S. Army Combat Capabilities Development Command Soldier Center (DEVCOM SC) will use your feedback to improve these items.
All responses are confidential.

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Demographics ver 5 - Multiservice

1. Select your branch

- ◯ Army
- ◯ USMC
- ◯ Navy
- ◯ Air Force
- ◯ Other (please specify): _____

2. Select your component:

- ◯ Active duty
- ◯ Reserves
- ◯ National Guard
- ◯ Other (please specify): _____

3. What unit are you with?

- ◯ 25 BSB
- ◯ 1-52 AVN
- ◯ 1/11 FSC
- ◯ JPMRC OPFOR
- ◯ 574 QM
- ◯ Other _____

4. What is your unit location?

- ◯ JBER
- ◯ Bridgeport
- ◯ Eielson
- ◯ Wainwright
- ◯ Other _____

5. Select your rank category:

- ◯ Enlisted
- ◯ Warrant Officer
- ◯ Officer

6. Select your rank grade: _____

7. Enter your primary MOS code: _____

8. How long have you served in the military? _____
Select the total number of years, or select "Less than one year".

If you selected anything other than “Less than one year”, skip to Q10.

9. [Less than one year of service] Enter your total months of service, under one year:
_____

10. Please indicate your sex:

- ◯ Male
- ◯ Female

COVERALLS

11. Did you use the 2-Piece Coverall?

◯ Yes
◯ No **— Skip to the HANDWEAR section.**

12. Which 2-Piece Coverall **COAT** were you issued (input the number on the garment label): _____

13. Which 2-Piece Coverall **TROUSER** were you issued (input the number on the garment label): _____

14. What platforms did you work with when wearing the coveralls? (select all that apply)

☐ Ground
☐ Air
☐ Other (please specify): _____

15. Which Cold Weather Clothing System did you use with the 2-Piece Coveralls? (select one)

◯ Gen III ECWCS
◯ ARTC (formerly CTAPS)
◯ Other (please specify): _____

16. What temperature ranges did you wear the 2-Piece Coverall in? (Select all that apply).

☐ -60 degrees or below
☐ -59 to -50 degrees
☐ -49 to -30 degrees
☐ -29 to -10 degrees
☐ -9 to 0 degrees
☐ 1 to 10 degrees
☐ 11 to 20 degrees
☐ 21 to 30 degrees
☐ 31 to 40 degrees
☐ 41 degrees or above

17. Approximately how many **hours per day** did you wear the 2-Piece Coverall, on average? (round to the closest hour)

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18. Approximately how many total **days** did you wear the 2-Piece Coverall?

_____

19. Did you wear the **Level 7 Extreme Cold Weather PARKA** (outmost insulating layer, "puffy" / "happy suit") under your 2-Piece Coverall **COAT**?

- ◯ Yes
- ◯ No

20. Did you wear the **Level 7 Extreme Cold Weather TROUSERS** (outmost insulating layer, "puffy" / "happy suit") under your 2-Piece Coverall **TROUSERS**?

- ◯ Yes
- ◯ No

Answer only if you selected "Yes" in Q19.

21. What is the **length** of the **Level 7 Extreme Cold Weather PARKA** ("puffy" / "happy suit") that you wore when evaluating the 2-Piece Coverall? Select all that apply if you wore more than one.

- ☐ Extra Short
- ☐ Short
- ☐ Regular
- ☐ Long
- ☐ Extra Long
- ☐ Other (please specify): _____
- ☐ Do not know

Answer only if you selected "Yes" in Q20.

22. What is the **length** of the **Level 7 Extreme Cold Weather TROUSER** ("puffy" / "happy suit") that you wore when evaluating the 2-Piece Coverall? Select all that apply if you wore more than one.

- ☐ Extra Short
- ☐ Short
- ☐ Regular
- ☐ Long
- ☐ Extra Long
- ☐ Other (please specify): _____
- ☐ Do not know

Answer only if you selected "Yes" in Q19.

23. Rate the **fit of the 2-Piece Coverall COAT** when worn atop the Level 7 Extreme Cold Weather **PARKA** ("puffy" / "happy suit"):

- ◯ Too tight (compresses the Level 7 puffy)
- ◯ Just right
- ◯ Too baggy/loose

Answer only if you selected "Yes" in Q20.

24. Rate the **fit of the 2-Piece Coverall TROUSERS** when worn atop the Level 7 Extreme Cold Weather **TROUSERS (**"puffy" / "happy suit"):

- ◯ Too tight (compresses the Level 7 puffy)
- ◯ Just right
- ◯ Too baggy/loose

25. Did you experience any durability issues with the 2-Piece coverall (top and/or bottom)?

- ◯ Yes
- ◯ No **— Skip to Q29.**

26. Please identify the type of durability issue(s) you experienced. (select all that apply)

- ☐ Seams opening
- ☐ Seam tape (on the inside) is peeling off
- ☐ Rips/holes in the material
- ☐ Wear/thinning of the material
- ☐ Hook and loop won't close
- ☐ Difficult /broken zipper
- ☐ Other (please specify): _____

27. Where on the garment did you experience these durability issues? (select all that apply)

- ☐ Hood
- ☐ Collar
- ☐ Sleeves
- ☐ Shoulders
- ☐ Coat pockets
- ☐ Front of coat
- ☐ Back of coat
- ☐ Belt/fly
- ☐ Stretch panel (on bottoms)
- ☐ Crotch
- ☐ Reinforced Seat
- ☐ Thighs
- ☐ Knees
- ☐ Lower leg
- ☐ Other (please specify): _____

28. Please describe what you were doing when the damage occurred.

29. Did your coveralls come into contact with any Petroleum, Oils, and Lubricants (POLs) during the evaluation?

- ◯ Yes
- ◯ No **— Skip to Q34.**

30. Please indicate which Petroleum, Oils, and Lubricants (POLs) the coveralls came into contact. (select all that apply)

- ☐ JP5
- ☐ JP8
- ☐ Kerosene
- ☐ Other (please specify): _____

31. How did the coveralls come into contact with POLs? (select all that apply)

- ☐ Spray
- ☐ Splash
- ☐ Steady Stream
- ☐ Knelt in fuel puddle
- ☐ Other (please specify): _____

32. Roughly how much POL came into contact with the coveralls?

- ◯ Up to 8oz (1 cup)
- ◯ 8oz to 16 oz (1-2 cups)
- ◯ 1 quart (2-4 cups)
- ◯ 1.5 quarts (4-6 cups)
- ◯ Half a gallon (8 cups)
- ◯ Up to a gallon
- ◯ More than a gallon
- ◯ Don't know

33. Did the coveralls provide adequate protection from the POLs?

- ◯ Yes
- ◯ No

34. Did you clean the coveralls?

- ◯ Yes
- ◯ No **— Skip to Q36.**

35. Did you have any problems after cleaning

- ◯ Yes (please describe) _____
- ◯ No

36. How much do you like or dislike the following features of the 2-Piece Coverall **COAT**:

	Dislike Very Much	Dislike Moderately	Dislike Slightly	Neither like nor Dislike	Like Slightly	Like Moderately	Like Very Much	N/A
Size of the hood	○	○	○	○	○	○	○	○
Hood sizing with cinches	○	○	○	○	○	○	○	○
Collar height	○	○	○	○	○	○	○	○
Interior zipper pocket (left side)	○	○	○	○	○	○	○	○
Outside zippered chest pocket	○	○	○	○	○	○	○	○
Waist cinch to keep wind out	○	○	○	○	○	○	○	○
Bottom hem cinch to keep wind out	○	○	○	○	○	○	○	○
Stretchable cuff	○	○	○	○	○	○	○	○
Pit zips	○	○	○	○	○	○	○	○
Shoulder reinforcement	○	○	○	○	○	○	○	○
Stowable hood	○	○	○	○	○	○	○	○
Flap opening chest & pen pockets	○	○	○	○	○	○	○	○

37. How much do you like or dislike the following features of the 2-Piece Coverall **TROUSER**:

	Dislike Very Much	Dislike Moderately	Dislike Slightly	Neither like nor Dislike	Like Slightly	Like Moderately	Like Very Much	N/A
Built-in belt	○	○	○	○	○	○	○	○
Zipper fly	○	○	○	○	○	○	○	○
Stretch panel on bottoms	○	○	○	○	○	○	○	○
Seat reinforcement	○	○	○	○	○	○	○	○
Size of cargo pocket	○	○	○	○	○	○	○	○
Lower leg zipper	○	○	○	○	○	○	○	○
Reinforced lower leg	○	○	○	○	○	○	○	○
Fit around boots	○	○	○	○	○	○	○	○

38. Were you **unable to perform** any of the following tasks when wearing the 2-Piece Coverall? (Select all that you were unable to perform, if any).

- ☐ Reaching up/overhead
- ☐ Reaching to the front
- ☐ Reaching to the side
- ☐ Bending over
- ☐ Squatting
- ☐ Taking a knee (on one knee)
- ☐ Getting on both knees
- ☐ Sitting
- ☐ Other (please specify): _____
- ☐ None of these - I was able to perform all tasks

39. How much do you like or dislike the 2-Piece Coverall, overall?

Like Very Much	Like Moderately	Like Slightly	Neither Like nor Dislike	Dislike Slightly	Dislike Moderately	Dislike Very Much
◯	◯	◯	◯	◯	◯	◯

40. Please comment about what you like and/or dislike about the 2-Piece Coverall:

41. Is the 2-Piece Coverall acceptable for mission use, as is?

- ◯ Yes, acceptable as is **— Skip to Q43.**
- ◯ No, it needs minor modifications
- ◯ No, it needs major modifications
- ◯ No, and it cannot be modified to meet my needs

42. Please explain what modifications are required to make the 2-Piece Coverall acceptable for mission use:

43. Please share any additional comments or recommendations about the 2-Piece Coverall or about being a fuel handler in an arctic environment:

HANDWEAR

44. Did you wear the hybrid handwear?

◯ Yes
◯ No **— Skip the remaining questions in this section.**

45. Which hybrid handwear were you issued **(LEFT glove/mitten)** (input the number on the stow-away straps) _____

46. Which hybrid handwear were you issued **(RIGHT glove/mitten)** (input the number on the stow-away straps) _____

47. What glove size(s) do you typically wear? Select all that apply.

☐ X-Small
☐ Small
☐ Medium
☐ Large
☐ X-Large

48. What size would call the hybrid handwear that you evaluated?

◯ X-Small
◯ Small
◯ Medium
◯ Large
◯ X-Large

49. Approximately how many **hours per day** did you wear the hybrid handwear, on average? (round to the closest hour) _____

50. Approximately how many **days** did you wear the hybrid handwear?
_____

51. Please rate the **length** of the hybrid handwear you evaluated on the following:

	Too Short	Just Right	Too Long
Finger length	◯	◯	◯
Palm length	◯	◯	◯
Mitten length	◯	◯	◯
Gauntlet length (portion that extends over arm)	◯	◯	◯

52. Please rate the **width** of the hybrid handwear you evaluated on the following:

	Too Narrow	Just Right	Too Wide
Finger width	◯	◯	◯
Palm width	◯	◯	◯
Mitten width	◯	◯	◯

53. Please rate the fit of the hybrid handwear you evaluated on the following:

	Too Small	Just Right	Too Big
Gauntlet opening (portion that extends over arm)	◯	◯	◯
Overall fit	◯	◯	◯

54. Did you experience any durability issues with your hybrid handwear?

◯ Yes
◯ No **— Skip to Q58.**

55. Please identify the type of durability issue(s) you experienced. (select all that apply)

☐ Rips/tears at the seams (seam is opening)
☐ Rips/holes in the brown shell material
☐ Cut or tears in the leather
☐ Leather has stiffened
☐ Black wrist strap pulled out
☐ Other (please specify): _____

56. Please identify where on the hybrid handwear you experienced an issue. (select all that apply)

- ☐ Glove (brown material)
- ☐ Glove leather
- ☐ Mitten (palm side)
- ☐ Gauntlet
- ☐ Wrist strap
- ☐ Stowaway strap
- ☐ Mitten (back side)
- ☐ Other (please specify): _____

57. Please describe what you were doing when the damage occurred.

_____

_____

_____

_____

_____

58. Did your hybrid handwear come into contact with any Petroleum, Oils, and Lubricants (POLs) during the evaluation?

- ◯ Yes
- ◯ No **— Skip to Q63.**

59. Please indicate which POLs the hybrid handwear came into contact. (select all that apply)

- ☐ JP5
- ☐ JP8
- ☐ Kerosene
- ☐ Other (please specify): _____

60. How did your hybrid handwear come into contact with the POLs? (select all that apply)

- ☐ Spray
- ☐ Splash
- ☐ Steady stream
- ☐ Immersed
- ☐ Other (please specify): _____

61. Roughly how much POL came into contact with the hybrid handwear?

- ◯ Up to 8oz (1 cup)
- ◯ 8oz to 16 oz (1-2 cups)
- ◯ 1 quart (2-4 cups)
- ◯ 1.5 quarts (4-6 cups)
- ◯ Half a gallon (8 cups)
- ◯ Up to a gallon
- ◯ More than a gallon
- ◯ Don't know

62. Did the hybrid handwear provide adequate protection from the POLs?

- ◯ Yes
- ◯ No (please explain): _____

63. Did you clean the hybrid handwear?

- ◯ Yes
- ◯ No **— Skip to Q65.**

64. Did you have any problems after cleaning

- ◯ Yes (please describe): _____
- ◯ No

65. Were you **unable to perform** any of the following tasks while wearing the glove **with the MITTEN OFF**? (Select all that you were unable to perform, if any).

- ☐ Ground refueling operations
- ☐ Aviation refueling operations
- ☐ Fuel sampling
- ☐ Carrying pipes
- ☐ Vehicle operations
- ☐ Using a manual gauge stick
- ☐ Hose operation
- ☐ Nozzle operation
- ☐ Using the pump control
- ☐ Opening/closing vehicle doors
- ☐ Asending/descending ladders
- ☐ Operating communications equipment
- ☐ Taking notes
- ☐ Repairing leaking/cut bags
- ☐ Other (please specify): _____
- ☐ None of these - Was able to perform all tasks

Answer only if you selected at least one response in Q65 and you did not select "None of these - Was able to perform all tasks" in Q65.

66. Please describe how you were unable to perform these tasks:

67. Were you **unable to perform** any of the following tasks while wearing the glove **with the MITTEN ON**? (Select all that you were unable to perform, if any).

- ☐ Ground refueling operations
- ☐ Aviation refueling operations
- ☐ Fuel sampling
- ☐ Carrying pipes
- ☐ Vehicle operations
- ☐ Using a manual gauge stick
- ☐ Hose operation
- ☐ Nozzle operation
- ☐ Using the pump control
- ☐ Opening/closing vehicle doors
- ☐ Asending/descending ladders
- ☐ Operating communications equipment
- ☐ Taking notes
- ☐ Repairing leaking/cut bags
- ☐ Other (please specify): _____
- ☐ None of these - Was able to perform all tasks

Answer only if you selected at least one response in Q67 and you did not select "None of these - Was able to perform all tasks" in Q67.

68. Please describe how you were unable to perform these tasks:

69. Please rate your level of satisfaction with the following features of the hybrid handwear **GLOVE (mitten off)**:

	Very Dissatisfied	Moderately Dissatisfied	Slightly Dissatisfied	Neither Satisfied nor Dissatisfied	Slightly Satisfied	Moderately Satisfied	Very Satisfied	N/A
Warmth when wearing glove only	○	○	○	○	○	○	○	○
Dexterity (ability to do tasks with fingers) with mitten off	○	○	○	○	○	○	○	○
Tactility (ability to feel objects) with mitten off	○	○	○	○	○	○	○	○
Protection from POLs with mitten off	○	○	○	○	○	○	○	○
Design of glove	○	○	○	○	○	○	○	○

70. Please rate your level of satisfaction with the following features of the hybrid handwear **MITTEN**:

	Very Dissatisfied	Moderately Dissatisfied	Slightly Dissatisfied	Neither Satisfied nor Dissatisfied	Slightly Satisfied	Moderately Satisfied	Very Satisfied	N/A
Protection from POLs with mitten on	○	○	○	○	○	○	○	○
Fit of mitten over glove	○	○	○	○	○	○	○	○
Mitten heater pocket location	○	○	○	○	○	○	○	○
Design of mitten	○	○	○	○	○	○	○	○

71. Please rate your level of satisfaction with the following **overall** hybrid handwear features:

	Very Dissatisfied	Moderately Dissatisfied	Slightly Dissatisfied	Neither Satisfied nor Dissatisfied	Slightly Satisfied	Moderately Satisfied	Very Satisfied	N/A
Overall warmth (glove plus mitten)	○	○	○	○	○	○	○	○
Comfort of the inner liner	○	○	○	○	○	○	○	○
Ability to fit into 2-piece coverall cargo pocket	○	○	○	○	○	○	○	○
Minimizing sweat build-up inside	○	○	○	○	○	○	○	○
Dry out time	○	○	○	○	○	○	○	○
Durability	○	○	○	○	○	○	○	○
Overall design	○	○	○	○	○	○	○	○
Overall fit	○	○	○	○	○	○	○	○
Overall performance	○	○	○	○	○	○	○	○

72. Please rate the effectiveness of the handwarmer pockets:

	Not at all Effective	Slightly Effective	Moderately Effective	Very Effective	Extremely Effective	N/A
Glove handwarmer pocket	○	○	○	○	○	○
Mitten handwarmer pocket	○	○	○	○	○	○

73. Please rate the ease of use of the hybrid handwear on the following:

	Very Difficult	Moderately Difficult	Slightly Difficult	Neither Easy nor Difficult	Slightly Easy	Moderately Easy	Very Easy	N/A
Ease of slipping gloved hand into mitten	◯	◯	◯	◯	◯	◯	◯	◯
Ease of removing mitten from gloved hand	◯	◯	◯	◯	◯	◯	◯	◯
Ease of securing the mitten when not in use (with the stowaway strap)	◯	◯	◯	◯	◯	◯	◯	◯
Ease of care for overall hybrid handwear	◯	◯	◯	◯	◯	◯	◯	◯

74. How much do you like or dislike the following features of the hybrid handwear?

	Dislike Very Much	Dislike Moderately	Dislike Slightly	Neither Like nor Dislike	Like Slightly	Like Moderately	Like Very Much
Leather grip	◯	◯	◯	◯	◯	◯	◯
Glove heater pocket	◯	◯	◯	◯	◯	◯	◯
Mitten heater pocket	◯	◯	◯	◯	◯	◯	◯
Pen pocket	◯	◯	◯	◯	◯	◯	◯
Black wrist strap / cinch	◯	◯	◯	◯	◯	◯	◯
Mitten stow-away strap	◯	◯	◯	◯	◯	◯	◯
Gauntlet cinch cord	◯	◯	◯	◯	◯	◯	◯
Black donning / lanyard loop	◯	◯	◯	◯	◯	◯	◯

75. How much do you like or dislike the hybrid handwear, overall?

Like Very Much	Like Moderately	Like Slightly	Neither Like nor Dislike	Dislike Slightly	Dislike Moderately	Dislike Very Much
◯	◯	◯	◯	◯	◯	◯

76. Please comment about what you like and/or dislike about the hybrid handwear:

_____

_____

_____

77. Is the hybrid handwear acceptable for mission use, as is?

- ◯ Yes, acceptable as is **— Skip to Q79.**
- ◯ No, it needs minor modifications
- ◯ No, it needs major modifications
- ◯ No, and it cannot be modified to meet my needs **— Skip to Q79.**

78. Please explain what modifications are required to make the hybrid handwear acceptable for mission use:

79. Please share any additional comments or recommendations about the hybrid handwear or about being a fuel handler in an arctic environment: